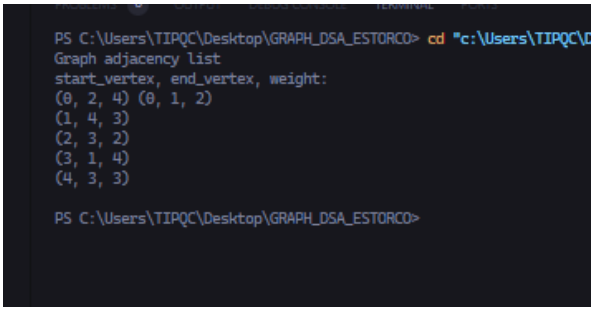
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COURSE CODE/TITLE:	CPE010 - Data Structure and Algorithm	**SECTION:**	CPE 21S2
DATE PERFORMED:	10/2/26	**DATE SUBMITTED:**	10/2/26
NAME:	Estorco, Clint Hoven A.	**INSTRUCTOR:**	Engr. Jimlord Quejado

Hands-On-Activity 10.1 Graph

1. PROCEDURE OUTPUT

ILO A
OUTPUT


```
PS C:\Users\TIPQC\Desktop\GRAPH_DSA_ESTORCO> cd "c:\Users\TIPQC\D
Graph adjacency list
start_vertex, end_vertex, weight:
(0, 2, 4) (0, 1, 2)
(1, 4, 3)
(2, 3, 2)
(3, 1, 4)
(4, 3, 3)

PS C:\Users\TIPQC\Desktop\GRAPH_DSA_ESTORCO>
```

ILO B.1
OUTPUT

```
4
2
PS C:\Users\TIPQC\Desktop\GRAPH_DSA_ESTORCO> cd "c:
1:      {2: 0}, {5: 0},
2:      {1: 0}, {5: 0}, {4: 0},
3:      {4: 0}, {7: 0},
4:      {2: 0}, {3: 0}, {5: 0}, {6: 0}, {8: 0},
5:      {1: 0}, {2: 0}, {4: 0}, {8: 0},
6:      {4: 0}, {7: 0}, {8: 0},
7:      {3: 0}, {6: 0},
8:      {4: 0}, {5: 0}, {6: 0},

DFS Order of vertices:
1
5
8
6
7
3
4
2
PS C:\Users\TIPQC\Desktop\GRAPH_DSA_ESTORCO>
* History restored
```

ILO B.2
OUTPUT

```
1:      {2: 2}, {5: 3},
2:      {1: 2}, {5: 5}, {4: 1},
3:      {4: 2}, {7: 3},
4:      {2: 1}, {3: 2}, {5: 2}, {6: 4}, {8: 5},
5:      {1: 3}, {2: 5}, {4: 2}, {8: 3},
6:      {4: 4}, {7: 4}, {8: 1},
7:      {3: 3}, {6: 4},
8:      {4: 5}, {5: 3}, {6: 1},
```

```
BFS Order of vertices:
1
2
5
4
8
3
6
7
```

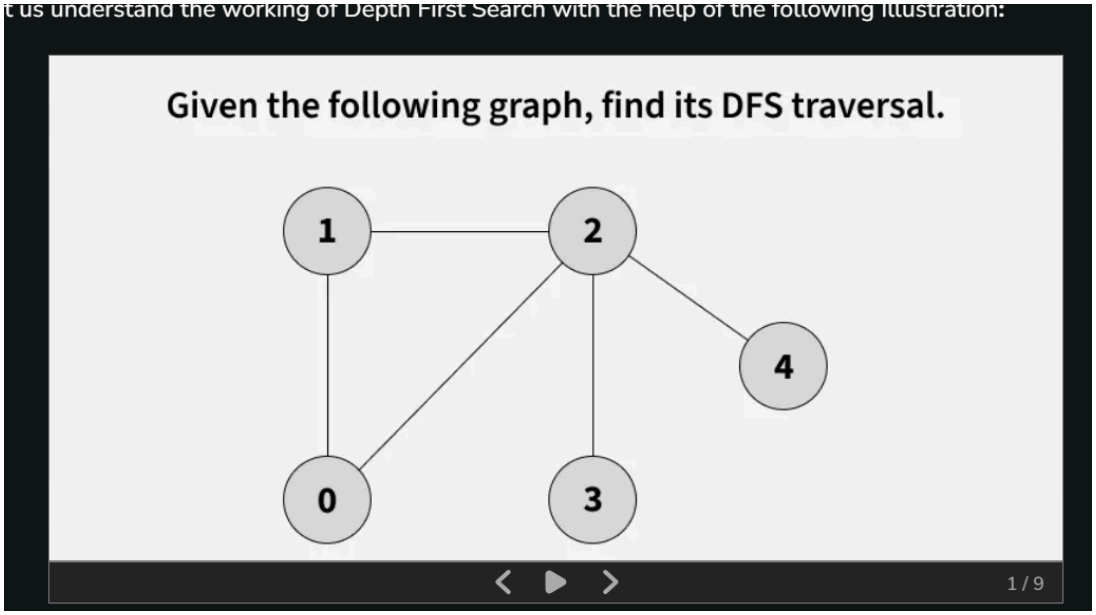
2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C

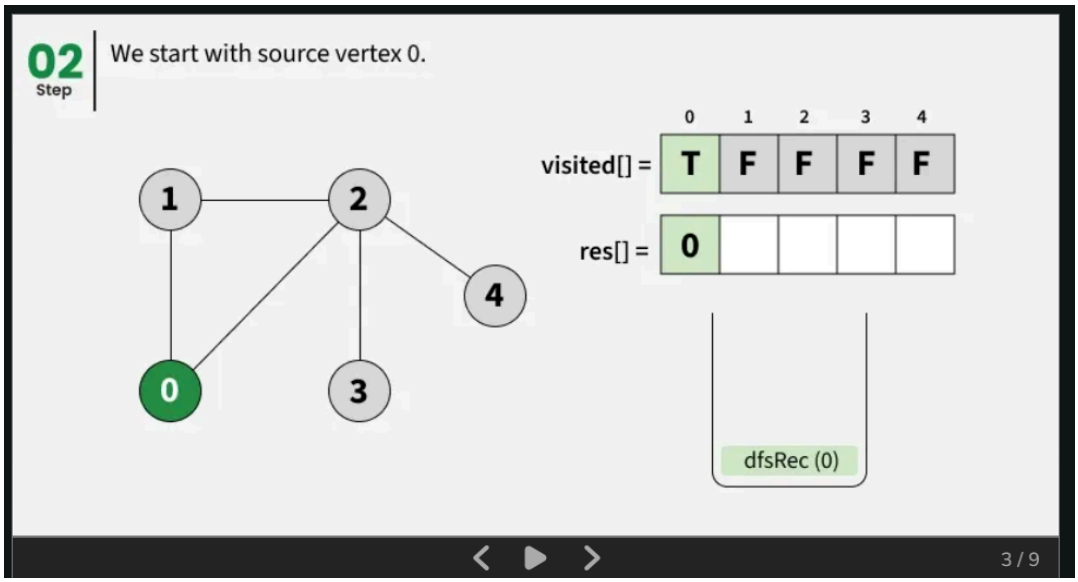
1. A person wants to visit different locations indicated on a map. He starts from one location (vertex) and wants to visit every vertex until it finishes from one vertex, backtracks, and then explore other vertex from the same vertex. Discuss which algorithm would be most helpful to accomplish this task.

- The appropriate algorithm for this scenario is going to be what is known as Depth-First Search (DFS) which will be based on a recursive backtracking method to explore all nodes. When applied to this problem, the traveler starts at a location and works as far along each path as he is able then reverses his direction to visit all the other vertices of the map.

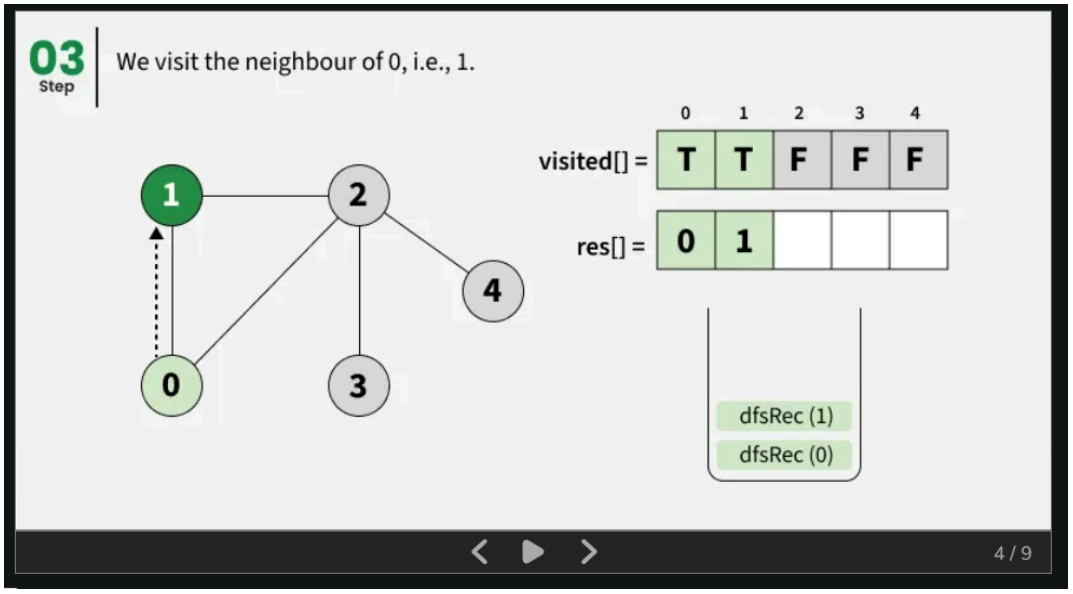
2. Identify the equivalent of DFS in traversal strategies for trees. To efficiently answer this question, provide a graphical comparison, examine pseudocode and code implementation.

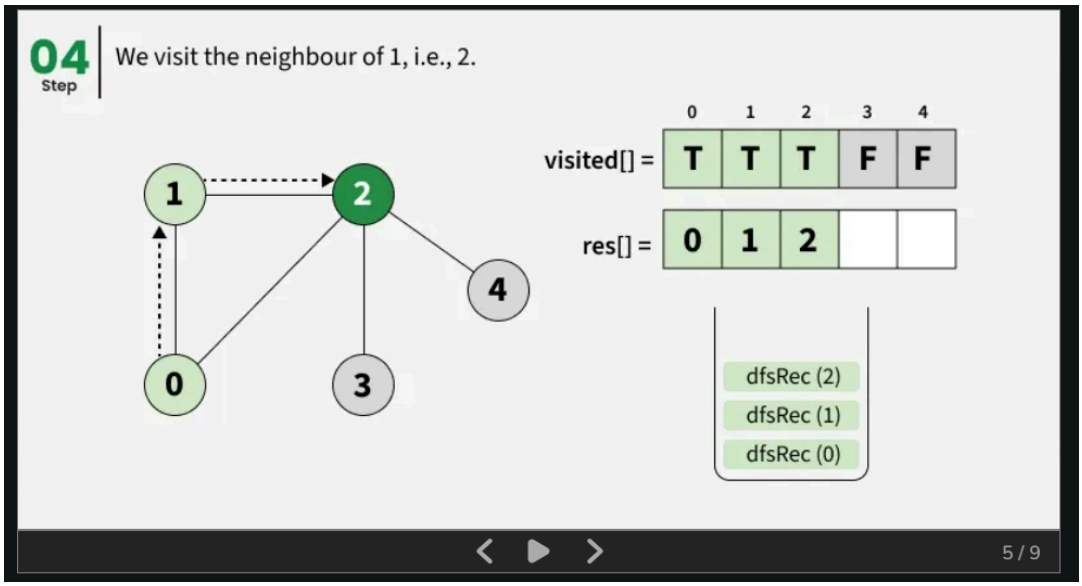
- 

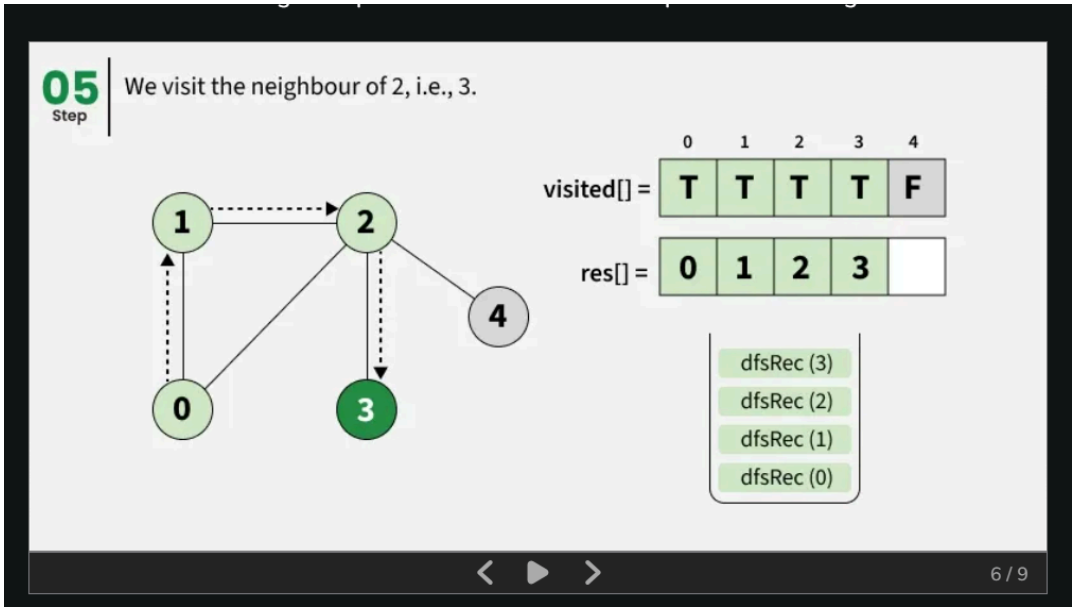
-

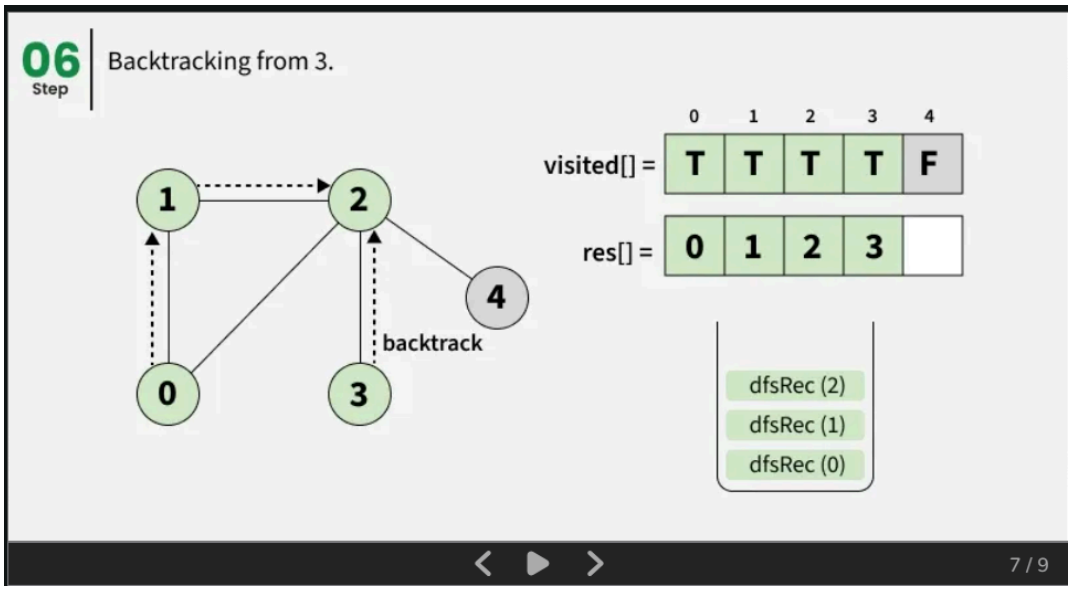

02
Step
We start with source vertex 0.
0 1 2 3 4
visited[] = T F F F F
res[] = 0
1
2
4
0
3
dfsRec (0)
3 / 9

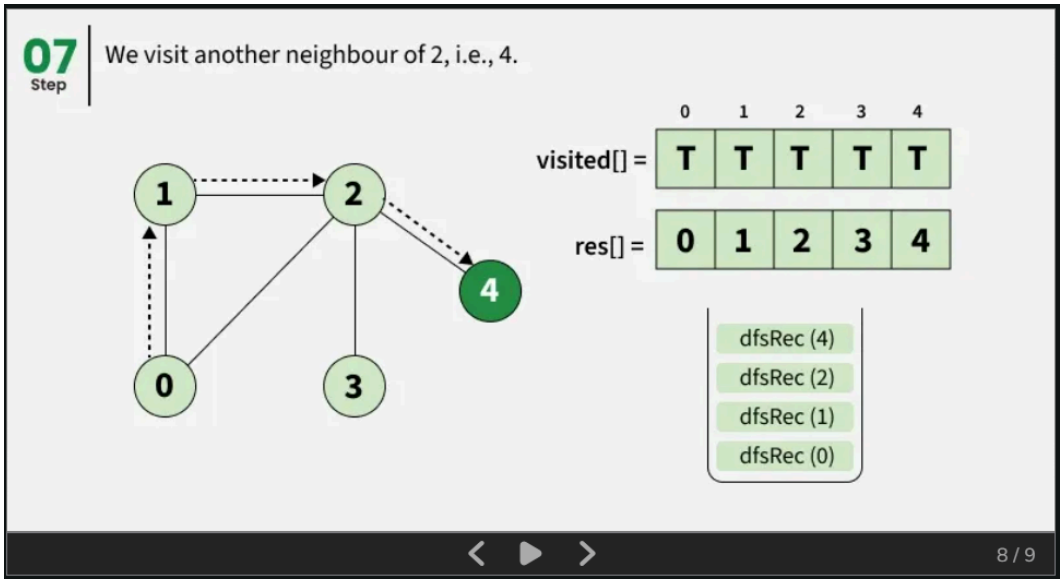
-

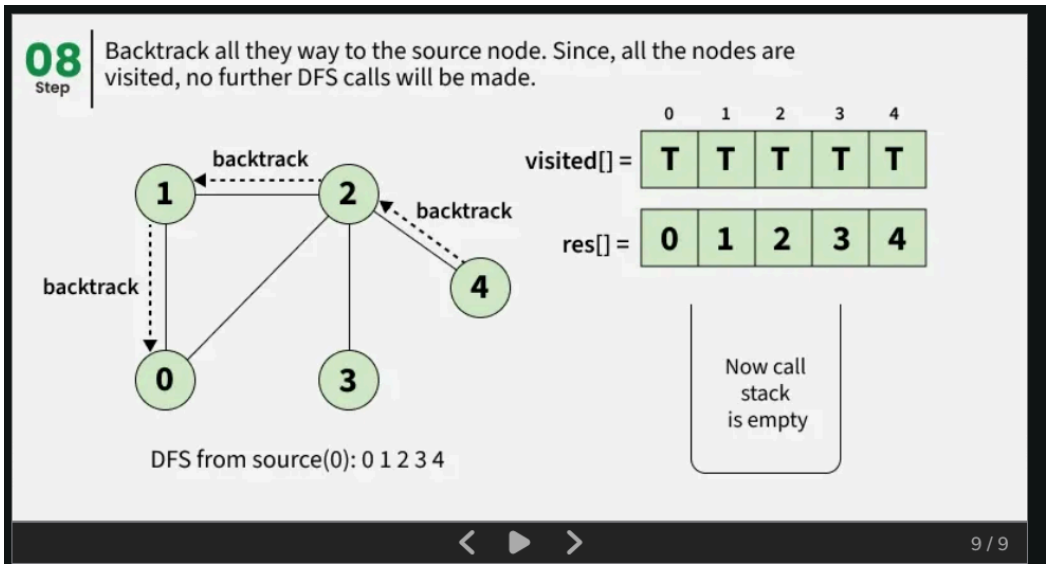

03
Step
We visit the neighbour of 0, i.e., 1.
0 1 2 3 4
visited[] = T T F F F
res[] = 0 1
1
2
4
0
3
dfsRec (1)
dfsRec (0)
4 / 9


04 Step
We visit the neighbour of 1, i.e., 2.
1
2
4
0
3
0 1 2 3 4
visited[] = T T T F F
res[] = 0 1 2
dfsRec (2)
dfsRec (1)
dfsRec (0)
5 / 9


05 Step
We visit the neighbour of 2, i.e., 3.
1
2
4
0
3
0 1 2 3 4
visited[] = T T T T F
res[] = 0 1 2 3
dfsRec (3)
dfsRec (2)
dfsRec (1)
dfsRec (0)
6 / 9


06 Step
Backtracking from 3.
0 1 2 3 4
visited[] = T T T T F
res[] = 0 1 2 3
1
2
4
backtrack
0
3
dfsRec (2)
dfsRec (1)
dfsRec (0)
7 / 9


07 Step
We visit another neighbour of 2, i.e., 4.
0 1 2 3 4
visited[] = T T T T T
res[] = 0 1 2 3 4
1
2
4
0
3
dfsRec (4)
dfsRec (2)
dfsRec (1)
dfsRec (0)
8 / 9

- 

3. In the performed code, what data structure is used to implement the Breadth First Search?

- Queue

4. How many times can a node be visited in the BFS?

- Once

REFERENCE:

https://www.geeksforgeeks.org/dsa/depth-first-search-or-dfs-for-a-graph/

Conclusion:

The methods used for this activity were successful in applying and analyzing the following graph traversal algorithms: Depth-First Search (DFS), Breadth-First Search (BFS). DFS was successful with the following methods: exhaustive path exploration, recursive backtracking, node visitation, and having a matching traversal of tree which is pre order. Developing layer by layer traversal using a Queue data structure is different compared to BFS where each node that can be reached was traversed once. Understanding these basic graph structures and methods of traversing them is essential to understanding a wide range of computer engineering problems that involve routes on networks, navigation through maps, and pathfinding in a maze.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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